

قسم علوم الحاسب

اسم الطالبة: الرقم الجامعي : مدة الاختبار: ساعة

اسم المقرر: بحوث العمليات

رقم المقرر ورمزه: Math329

المستوى: السابع

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الاختبار الفصلي للفصل الدراسي الأول (٤٥١) للعام الجامعي 1444-1445هـ

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استعيني بالله وأجيبني على جميع الأسئلة التالية

السؤال الأول: اختاري الإجابة الصحيحة من بين الإجابات الواردة في كل فقرة

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1. Operations research is the application of _____ methods to arrive at the optimal Solutions to the problems.
A. economical
B. scientific
C. a and b both
D. artistic
2. In operations research, the -----are prepared for situations.
A. mathematical models
B. physical models diagrammatic
C. diagrammatic models
3. Which of the following is not the phase of OR methodology?
A. Formulating a problem
B. Constructing a model
C. Establishing controls
D. Controlling the environment
4. Which technique is used in finding a solution for optimizing a given objective, such as profit maximization or cost reduction under certain constraints?
A. Quailing Theory
B. Waiting Line
C. Both A and B
D. Linear Programming
5. What is the objective function in linear programming problems?
A. A constraint for available resource
B. An objective for research and development of a company
C. A linear function in an optimization problem
D. A set of non-negativity conditions

6. Which statement characterizes standard form of a linear programming problem?
- A. Constraints are given by inequalities of any type
 - B. Constraints are given by a set of linear equations
 - C. Constraints are given only by inequalities of $\geq$ type
 - D. Constraints are given only by inequalities of $\leq$ type
7. In graphical representation the bounded region is known as _____ region.
- A. Solution
 - B. basic solution
 - C. feasible solution
 - D. optimal
8. Graphical optimal value for Z can be obtained from
- A. Corner points of feasible region
 - B. Both a and c
 - C. corner points of the solution region
 - D. none of the above
9. In assignment problem of maximization, the objective is to maximise
- A. Profit
 - B. optimization
 - C. cost
 - D. None of the above
10. OR has a characteristics that it is done by a team of
- A. Scientists
 - B. Mathematicians
 - C. Academics
 - D. All of the above
11. What have been constructed from OR problems and methods for solving the models that are available in many cases?
- A. Scientific Models
 - B. Algorithms
 - C. Mathematical Models
 - D. None of the above
12. Operations Research study generally involves how many phases ?
- A. Three
 - B. Four
 - C. Five
 - D. Two
13. As for maximization in assignment problem, the objective is to maximize the -----
- A. Profit
 - B. optimization
 - C. cost
 - D. None of the above

1-The demand function for a good is given as $Q = 130 - 10P$. Fixed costs associated with producing that good are €60 and each unit produced costs an extra €4.

- i) Obtain an expression for total revenue and total costs in terms of Q
- ii) For what values of Q does the firm break even
- iii) Obtain an expression for profit in terms of Q and sketch its graph
- iv) Use the graph to confirm your answer to (ii) and to estimate maximum profit and the level of output for which profit is maximized.

الحل :

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بالتوفيق والنجاح